

Deep Explanations — Biology Chapter 7: Metabolism

The "why does this work?" companion. Read this when a concept in [notes.md](#) doesn't quite click. Take your time — these are meant to be read slowly, one topic at a time.

Topic 1 — Metabolism and Its Types: Life Is Chemistry

Think about what you did this morning. You woke up, opened your eyes, stretched, maybe yawned. All of that — the muscle contractions, the nerve impulses, the slow warming of your body — required energy. And all that energy came from the food you ate yesterday.

But how does a piece of roti become a muscle contraction? How does a spoonful of rice become a thought?

The answer is metabolism: the sum of every chemical reaction happening inside every cell of your body, every second of every day. The word itself comes from the Greek *metabole*, meaning "change." And that's exactly what metabolism is — it's life constantly changing one thing into another.

Catabolism — the demolition crew

Imagine a team of workers tearing down an old building. They smash walls, break concrete, rip out wiring. The building becomes a pile of smaller pieces. That's catabolism in a nutshell.

Catabolic reactions **break complex molecules down into simpler ones**. When you digest food, large protein molecules are broken into amino acids, complex carbohydrates are broken into glucose, and fats are broken into fatty acids. When your cells respire, glucose is broken down further still — all the way to carbon dioxide and water.

The critical thing about catabolism: it **releases energy**. Every time you break a chemical bond (the right kind of bond), energy comes out. That released energy is what powers your life.

Anabolism — the construction crew

Now imagine the opposite team — the builders. They take bricks and cement and wood and construct a new building. That's anabolism.

Anabolic reactions **build complex molecules from simpler ones**. When a plant takes carbon dioxide and water and builds glucose during photosynthesis, that's anabolism. When your muscle cells take amino acids and build muscle proteins, that's anabolism. When your liver converts glucose to glycogen to store for later, that's anabolism.

The critical thing about anabolism: it **consumes energy**. You can't build without putting in work. That's why athletes need to eat — they're fueling the anabolism that builds muscle.

The beautiful partnership

Catabolism and anabolism are not enemies — they're partners. Catabolism generates energy; anabolism uses that energy. Your body is a constant cycle of tearing down (food, old proteins) and building up (new proteins, energy stores, cell parts).

Stop catabolism and you run out of energy. Stop anabolism and your cells fall apart. Both must happen, all the time.

That's why metabolism is the right word for the whole thing. Life isn't static — it's a constant, controlled, purposeful chemical change.

Topic 2 — Enzymes: The Protein Machines That Run Everything

Here's a problem. The chemical reactions your body needs don't just happen on their own at room temperature. If you left a bowl of glucose sitting in air, it would take years to break down spontaneously. But your cells break down glucose in milliseconds.

How? With enzymes.

What enzymes actually are

Enzymes are proteins — and not just any proteins, but **globular proteins**: chains of amino acids folded into a precise 3D shape. That shape isn't decorative. It's functional. The specific 3D shape of an enzyme is what allows it to do its job.

Most enzymes are proteins, though there are a few rare exceptions called **ribozymes** — enzymes made of RNA instead of amino acids. But for this chapter, think "enzyme = protein."

Why enzymes are different from ordinary catalysts

A chemistry teacher might put a metal catalyst in a reaction to speed it up — like iron in the Haber process for making ammonia. Enzymes do the same job, but they're alive (well, they're made by living cells), and they're staggeringly more specific.

An iron catalyst will speed up many different reactions. Amylase — the enzyme in your saliva — will speed up exactly ONE reaction: the breakdown of starch. It won't touch proteins, won't touch fats, won't even break down a slightly different carbohydrate. It's like having a locksmith who can only open one specific lock. One lock, millions of times per second, perfectly every time.

Enzymes don't start reactions — they speed them up

This is a subtle but important point. A reaction can't start if there's a barrier called the **activation energy** — the minimum energy the molecules need to collide and react. Enzymes lower this barrier. They make it easier for the reaction to happen.

But if the reaction is thermodynamically impossible (if the universe simply doesn't allow it), no enzyme can make it happen. Enzymes speed up possible reactions; they don't create impossible ones.

The cofactor story

Many enzymes need a helper to function. These helpers are called **cofactors**. There are three kinds:

Activators are simple inorganic ions — like zinc (Zn^{2+}) or chloride. They attach loosely and help position the substrate or stabilize the enzyme's shape.

Prosthetic groups are organic molecules that permanently fuse to the enzyme. FAD (Flavin Adenine Dinucleotide) is a famous example — it's stuck to the enzyme it works with and never lets go.

Coenzymes are organic molecules that attach and detach as needed. NAD (Nicotinamide Adenine Dinucleotide) is one — it carries electrons from one reaction to another, acting like a shuttle between metabolic pathways. Vitamins B and C are coenzymes (which is one reason vitamin deficiencies cause metabolic problems — you're running an enzyme without its cofactor).

Topic 3 — Mechanism of Enzyme Action: The Lock and the Key

The most famous image in enzyme biology is the "lock and key" model, proposed by the German chemist Emil Fischer in 1894. He imagined the enzyme as a lock and the substrate as a key — only the right key (the right shape) fits the lock.

This model was a brilliant start. But later work showed it wasn't quite right — the lock doesn't stay rigid. When the right substrate approaches, the enzyme actually **changes shape slightly** to fit around the substrate more snugly. This is called the **induced fit model**, like a glove that molds to your hand.

The active site — the most important tiny spot on Earth

Every enzyme has a region called the **active site** — a small, specifically shaped cavity on the enzyme's surface where the substrate binds. The active site is usually only a tiny fraction of the entire enzyme — but it's where everything happens.

The active site has a characteristic shape (complementary to the substrate) and a particular charge distribution. Both shape AND charge must match for the substrate to bind. This is why each enzyme is so specific: if the shape or charge is wrong, the substrate simply won't stick.

The ES complex — when chemistry happens

When the right substrate molecule floats close enough to an enzyme's active site, it binds. The resulting structure —

enzyme plus substrate, held together — is called the **Enzyme-Substrate (ES) Complex**. This is the moment the catalysis begins.

Inside the ES complex, the enzyme manipulates the substrate: it might bend a bond, transfer a group of atoms, or bring two molecules together. The actual mechanism varies by enzyme, but the result is always the same: the substrate is transformed into a **product**.

The ES complex becomes an **Enzyme-Product (EP) Complex** — the enzyme still holding what was the substrate, now transformed into products. Then the EP complex falls apart: the products float away, and the enzyme is left exactly as it started — unchanged, ready to do it again.

This is why enzymes are needed only in tiny amounts. One enzyme molecule can catalyze the same reaction millions of times without being used up.

Topic 4 — Factors Affecting Enzyme Activity: When Machines Break

Every machine has an operating range. Run a car engine at -40°C and it won't start. Run it at 300°C and it melts. Enzymes are the same.

Temperature — the Goldilocks problem

Enzymes work best at their **optimum temperature** — for human enzymes, that's about 37°C . Not coincidentally, that's your body temperature. Evolution has tuned our enzymes for the environment they live in.

Below the optimum temperature, enzymes are sluggish. The molecules don't have enough energy to collide and bind properly. The enzyme is still intact — it just doesn't work fast. Importantly, if you warm it back up, it works again. Cold doesn't destroy an enzyme; it just slows it down.

Above the optimum temperature, something far worse happens. The heat causes the protein to vibrate so violently that the carefully folded 3D structure collapses. The active site is destroyed. The enzyme is **denatured** — and unlike being cold, denaturation is usually irreversible. You can't refold a cooked egg white. You can't un-denature an enzyme.

pH — the acid-base problem

Every enzyme has an **optimum pH** too. Changes in pH affect the charges on the amino acids in the active site. Change those charges, and the substrate no longer fits.

Pepsin — your stomach enzyme that digests protein — has an optimum pH of about 2. That's highly acidic. Your stomach produces HCl specifically to create that acid environment, because pepsin won't work otherwise.

Trypsin — the same job (protein digestion), but in the small intestine — has an optimum pH of about 8. That's alkaline. When food leaves the stomach, your intestine adds bicarbonate to neutralize the acid, partly to protect the intestinal walls but also to give trypsin the pH it needs.

So when your teacher asks "why does the pH of the stomach matter?" — the real answer is: because pepsin is extremely fussy about its working environment.

Substrate concentration — traffic jams at the active site

Imagine a factory with 10 workers (enzymes) and a conveyor belt bringing in raw materials (substrate). If there's very little material on the belt, workers are idle — adding more material speeds things up. But once all 10 workers are fully occupied, adding more material to the belt just piles up in a queue. The workers can't go faster.

This is exactly what happens with substrate concentration. Up to a point, more substrate means faster reaction. But once every enzyme's active site is occupied — a condition called **saturation** — adding more substrate does nothing. The rate is limited by the number of enzyme molecules, not the amount of substrate.

Topic 5 — Enzyme Inhibition: Sabotage from Inside and Outside

Your body doesn't just need enzymes to run — it also needs to be able to turn them off. Imagine if you could never stop digesting. Or if a metabolic pathway kept running even after the cell had all the product it needed. Enzyme inhibition is

how the cell applies the brakes.

But inhibition can also come from outside — from poisons, drugs, antibiotics. Understanding how inhibitors work explains why certain poisons kill, and how certain medicines treat disease.

Competitive inhibition — the impostor

A competitive inhibitor is a molecule that is structurally similar to the substrate. It's like an impostor key — shaped just enough like the real key to fit in the lock, but not the right key, so it doesn't open the door.

The inhibitor occupies the active site. While it's sitting there, the real substrate can't bind. No substrate binding means no ES complex, no product.

But here's the key insight: this is a **competition**. Both the inhibitor and the real substrate are fighting for the same active site. If you flood the reaction with more substrate, the probability that the active site will be occupied by a real substrate (rather than the impostor) increases. At high enough substrate concentrations, you can out-compete the inhibitor and restore most enzyme activity.

This is why competitive inhibition is called **reversible**.

Non-competitive inhibition — the wrench in the gears

A non-competitive inhibitor doesn't look like the substrate at all. It doesn't bother with the active site. Instead, it binds to a completely different part of the enzyme called the **allosteric site** (from the Greek *allos* = other, *stereos* = solid/shape).

When something binds to the allosteric site, it causes the entire enzyme to change shape — including the active site. The active site is now distorted. The substrate can still try to bind, but it can't properly, because the shape is wrong.

Unlike competitive inhibition, adding more substrate doesn't help — the active site is damaged, not just blocked. Non-competitive inhibition is generally irreversible (or very difficult to reverse).

Cyanide — the famous poison — is a non-competitive inhibitor of a critical enzyme in the electron transport chain. Block that enzyme, and the entire aerobic respiration pathway stops. No ATP. No life.

Topic 6 — ATP: Why Life Uses a Universal Energy Coin

Life needs energy. But the problem with energy is that it comes in many different forms — sunlight, chemical bonds in glucose, heat — and different cellular machines need different amounts and types of energy. How does the cell manage this diversity?

The answer is brutally elegant: **convert everything into one standard currency, and then spend that currency as needed.**

That currency is ATP: Adenosine Triphosphate.

Why a triphosphate?

ATP has three phosphate groups linked in a chain. The bonds between the phosphate groups — especially the bond linking the second and third phosphate — are described as "high-energy bonds." When this bond is broken (by adding water — a process called hydrolysis), energy is released. About 30–32 kJ per mole, which is exactly the right amount for most cellular reactions.

ATP becomes ADP (Adenosine Diphosphate) and a loose phosphate group. The energy released powers whatever the cell needs: a muscle fiber contracting, a nerve impulse firing, a protein being synthesized.

The ATP-ADP cycle — a rechargeable battery

ADP doesn't stay as ADP. The cell constantly recharges it back into ATP by using energy from catabolism. This is called **phosphorylation** — adding that third phosphate back onto ADP.

During aerobic respiration, one glucose molecule produces about 36 ATP molecules. During photosynthesis, light energy from the sun recharges ADP back to ATP. Both processes are doing the same fundamental thing: capturing energy and storing it in those high-energy phosphate bonds.

Then the cell spends the ATP as needed. The cycle runs constantly — your cells hydrolyze and re-synthesize around 100-

150 moles of ATP every single day. The same ATP molecule might be recycled hundreds of times.

Topic 7 — Photosynthesis and Respiration: The Grand Cycle of Energy

Photosynthesis — the sun-eating trick

Imagine being able to eat sunlight. That's essentially what a plant does. Photosynthesis is the process by which plants (and algae and cyanobacteria) capture light energy and store it in chemical form as glucose.

The raw materials are humble: carbon dioxide from the air and water from the soil. The energy source is sunlight, captured by a green pigment called **chlorophyll**. The products are glucose (energy stored in chemical bonds) and oxygen (a waste product released into the air).

The process happens in two phases inside the chloroplast:

Phase 1 — Light-dependent reactions (in the grana/thylakoids): Chlorophyll absorbs light. This energy is used to split water molecules (photolysis), releasing oxygen and loading energy onto electron carriers. The energy ends up stored in ATP and NADPH — two energy-carrying molecules that will power phase 2.

Phase 2 — Light-independent reactions/Calvin cycle (in the stroma): Using the ATP and NADPH from phase 1, carbon dioxide is grabbed from the air and attached to a 5-carbon molecule (RuBP) to start building sugar. Through a cycle of reactions (the Calvin cycle, named after Melvin Calvin), those carbon atoms are slowly assembled into a 3-carbon sugar called G3P, which is the building block for glucose.

Why is it called "light independent"? Because the Calvin cycle doesn't directly use light. It uses the ATP and NADPH that were *made* by the light reactions. It could theoretically run at night — if you gave it a supply of ATP and NADPH. In practice, it runs during the day because that's when the light reactions are producing those molecules.

Aerobic Respiration — spending the energy glucose stored

When a cell needs energy, it burns glucose — carefully, in controlled steps. The whole process of aerobic respiration happens in four stages:

Stage 1 — Glycolysis (cytoplasm): The word comes from *glyco* (sugar) + *lysis* (splitting). One glucose molecule (6 carbons) is split in half to produce two pyruvic acid molecules (3 carbons each). A small return: 2 ATP and 2 NADH. This happens outside the mitochondria, in the cytoplasm.

Stage 2 — Formation of Acetyl CoA (mitochondria): Each pyruvic acid is oxidized. A carbon is removed as CO₂, and the remaining 2-carbon "acetyl" group attaches to Coenzyme A to form Acetyl CoA. More NADH is produced.

Stage 3 — Krebs Cycle (mitochondrial matrix): Acetyl CoA enters the cycle. The 2-carbon acetyl group is fed into a series of reactions that release both carbons as CO₂ while generating electron carriers (NADH and FADH₂) and a small direct ATP yield. The cycle turns twice per glucose (once per pyruvic acid).

Stage 4 — Electron Transport Chain (inner mitochondrial membrane): The NADH and FADH₂ are electron-rich — they carry the electrons harvested from glucose's bonds. These electrons flow down the ETC like water down a series of waterwheels, and each fall generates energy to pump protons and ultimately synthesize ATP. At the end of the chain, oxygen accepts the electrons (that's why you breathe oxygen — it's the final "drain" for the electrons) and combines with hydrogen to make water.

The total: 36 ATP per glucose molecule. Compare that to the 2 ATP from anaerobic — aerobic respiration is 18 times more efficient.

Anaerobic respiration — when oxygen isn't available

Sometimes the cell can't get enough oxygen — during intense exercise, or in environments where oxygen is absent. In those cases, glycolysis still runs (it doesn't need oxygen), but the pyruvic acid can't enter the Krebs cycle. Instead, it takes a shortcut:

In yeast and some bacteria: pyruvic acid → ethyl alcohol + CO₂ (**alcoholic fermentation**). This is how beer and bread are made.

In your muscle cells during a sprint: pyruvic acid → lactic acid (**lactic acid fermentation**). The burning sensation in your muscles after intense exercise is lactic acid accumulating. When you stop and catch your breath, oxygen arrives and clears the lactic acid.

Anaerobic respiration produces only 2 ATP — enough to survive a sprint, but not enough to sustain life long-term.

Photosynthesis and respiration — two sides of the same coin

Here's a beautiful symmetry: the equation for photosynthesis and the equation for aerobic respiration are exact reverses of each other.

- **Photosynthesis:** $6\text{CO}_2 + 12\text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2$
- **Aerobic respiration:** $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{energy}$

Plants capture energy and store it in glucose. Animals (and plants at night) release that energy from glucose. The oxygen plants release is the same oxygen animals breathe. The CO_2 animals exhale is the same CO_2 plants use.

Life on Earth is one giant, interlocking metabolic cycle. And understanding it starts right here.

Putting it all together

Look back at all 7 topics and notice the thread:

1. Life runs on chemical reactions (metabolism).
2. Those reactions need controllers (enzymes).
3. Those controllers need the right conditions (factors).
4. They can be switched off when needed (inhibition).
5. They all ultimately run on one energy currency (ATP).
6. Plants make that currency from sunlight (photosynthesis).
7. All living things spend it by burning glucose (respiration).

This whole chapter is one single story — **how life captures, stores, and spends energy** — told from seven different angles.

Once that clicks, everything in this chapter becomes easy.